EE2001 Electrical Circuits- I:

Continuation of the Topic:

Response of First-Order RL and RC Circuits

19 December 2023

Outline



Continuation of the topic:

4. The Step and the Total Responses of RL and RC Circuits

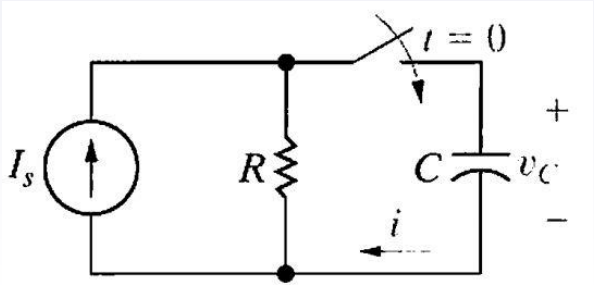
5. Sequential Switching

6. Unbounded Response

The Step and the Total Responses of RL and RC Circuits

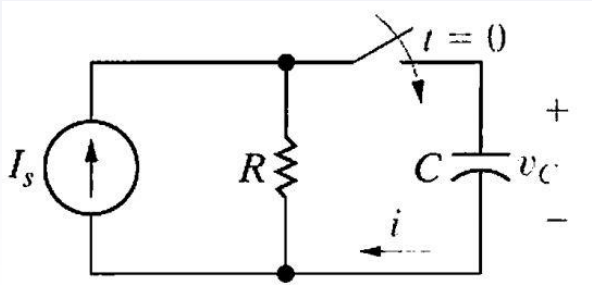
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- Consider the circuit



- If $v(0) = 0$, step response.
- If $v(0) = V_0 \neq 0$, total response.
- We can write a loop (KCL) equation to solve for the response: $-I_s + C\frac{dv_C}{dt} + \frac{v_C}{R} = 0$

The Step and the Total Responses of RL and RC Circuits



- If $v(0) = 0$ (step response)

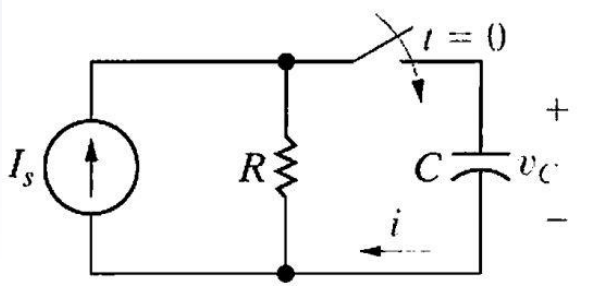
$$v_C(t) = I_s R - I_s R e^{-\frac{t}{RC}}$$

- If $v(0) = V_0 \neq 0$ (total response)

$$v_C(t) = I_s R + (V_0 - I_s R) e^{-\frac{t}{RC}}$$

The Step and the Total Responses of RL and RC Circuits

- Consider the total response



$$v_C(t) = I_s R + (V_0 - I_s R)e^{-\frac{t}{RC}}$$

$$i(t) = (I_s - \frac{V_0}{R})e^{-\frac{t}{RC}}$$

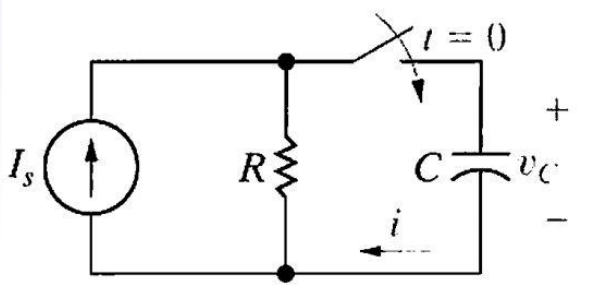
- It can be rewritten in the form

$$v_C(t) = \underbrace{V_0 e^{-\frac{t}{RC}}}_{\text{natural response}} + \underbrace{I_s R\left(1 - e^{-\frac{t}{RC}}\right)}_{\text{step response}}$$

- The first part is the response due to the initial conditions (natural response)
- The second part is the response due to the input (step response)

The Step and the Total Responses of RL and RC Circuits

- Consider the step response



$$v_C(t) = I_s R - I_s R e^{-\frac{t}{RC}}$$

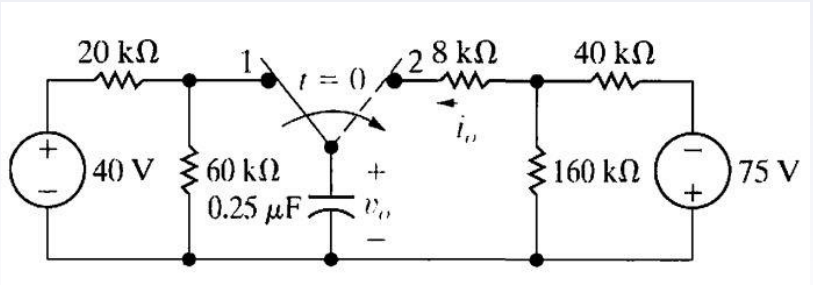
$$i(t) = I_s e^{-\frac{t}{RC}}$$

- The plot of the response is in the form

The Step and the Total Responses of RL and RC Circuits

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- Example-6: The switch in the given circuit has been in position 1 for a long time. At $t = 0$, the switch moves to position 2. Find



a) $v_0(t)$ for $t \geq 0$

b) $i_0(t)$ for $t \geq 0^+$

Solution: a) The switch has been in position 1 for a long while, so considering 40 V source divided across the 20 kΩ and 60 kΩ resistors, the initial value of v_0

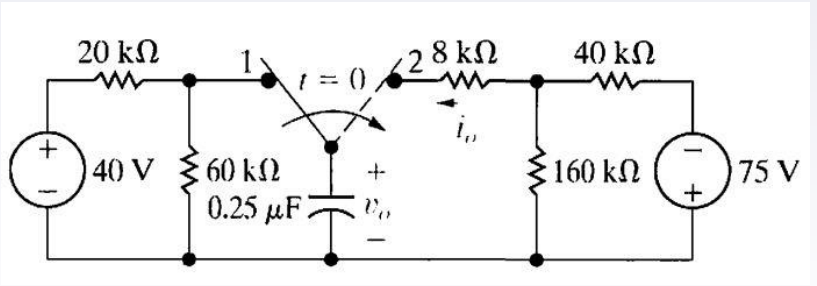
$$v_0(0) = V_0 = \frac{60\times10^3}{(20+40)\times10^3}(40) = 30\ V$$

We will find the Norton equivalent with respect to the terminals of the capacitor for $t \geq 0$.

We begin by computing the open-circuit voltage, which is -75 V source divided across 40 kΩ and 160 kΩ resistors: $V_{oc} = \frac{160\times10^3}{(40+160)\times10^3}(-75) = -60\ V$

The Step and the Total Responses of RL and RC Circuits

- Example-6: The switch in the given circuit has been in position 1 for a long time. At $t = 0$, the switch moves to position 2. Find



a) $v_0(t)$ for $t \geq 0$

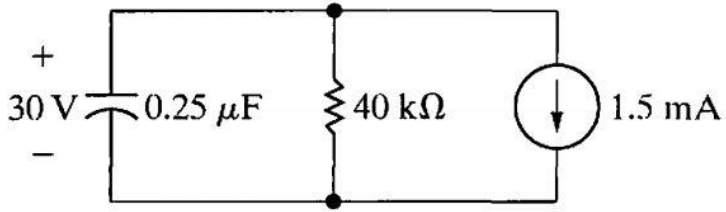
b) $i_0(t)$ for $t \geq 0^+$

Solution: a) Next, we calculate the Thevenin resistance, as seen to the right of the capacitor, by shorting the -75V source and making series and parallel combinations of the resistors:

$$R_{Th} = 8000 + (40000||160000) = 40\ k\Omega$$

The value of the Norton current source is the ratio of the open-circuit voltage to the Thevenin resistance $I_s = -\frac{60}{40\times10^3} = -1.5\ mA$

The resulting norton equivalent circuit:

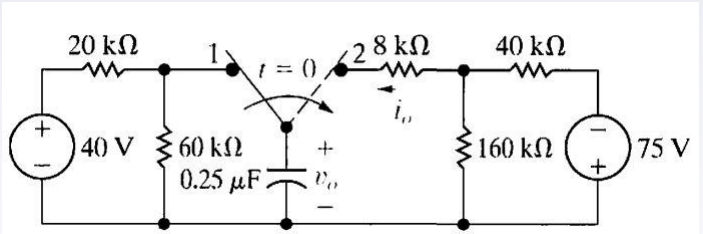


$I_sR = -60\ V$ and $RC = 10\ ms$

The Step and the Total Responses of RL and RC Circuits

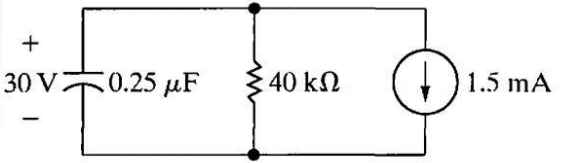
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- Example-6: The switch in the given circuit has been in position 1 for a long time. At $t = 0$, the switch moves to position 2. Find



a) $v_0(t)$ for $t \geq 0$

b) $i_0(t)$ for $t \geq 0^+$



Solution: a) We have already noted that $v_0(0) = 30\ V$ so the solution for $v_0(t)$ is

$$v_0(t) = -60 + \big(30 - (-60)\big)e^{-100t}$$
$$= -60 + 90e^{-100t}\ V, \qquad t \geq 0$$

b) Noting that $I_s = -1.5\ mA$ and $\frac{V_0}{R} = \left(\frac{30}{40}\right) \times 10^{-3} = 0.75\ mA$

$$i_0 = -2.25e^{-100t}\ mA, \qquad t \geq 0^+$$

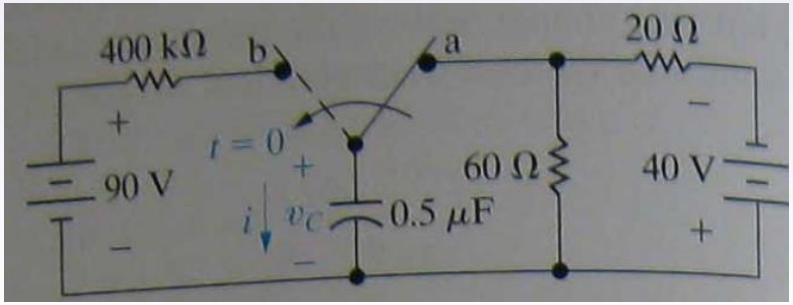
We can check the consistency of the solutions for $v_0(t)$ and $i_0(t)$ by

$$i_0(t) = C\frac{dv_0}{dt} = (0.25 \times 10^{-6})(-9000e^{-100t}) = -2.25e^{-100t}\ mA$$

The Step and the Total Responses of RL and RC Circuits

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- Example-7: The switch in the given circuit has been in position a for a long time. At $t = 0$, the switch moves to position b.



a) What is the initial value of v_C?

b) What is the final value of v_C?

Solution: a) The switch has been in position a for a long time, so the capacitor looks like an open circuit. Therefore, the voltage across the capacitor is the voltage across the 60 Ω resistor. From the voltage divider rule, the voltage across the 60 Ω resistor is

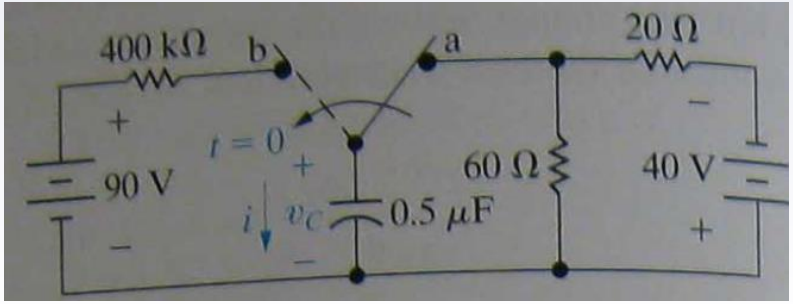
$$v_C(0) = -40 \times \left[\frac{60}{60 + 20}\right] = -30\ V$$

b) After the switch has been in position b for a long time, the capacitor will look like an open circuit in terms of the 90 V source. Thus, the final value of the capacitor voltage is + 90 V.

The Step and the Total Responses of RL and RC Circuits

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- Example-7: The switch in the given circuit has been in position a for a long time. At $t = 0$, the switch moves to position b.



c) What is the time constant of the circuit when the switch is in position b?

d) What is the expression for $v_C(t)$ when $t \geq 0$?

e) What is the expression for $i(t)$ when $t \geq 0^+$?

Solution: c) The time constant is

$$\tau = RC = (400 \times 10^3)(0.5 \times 10^{-6}) = 0.2\ s$$

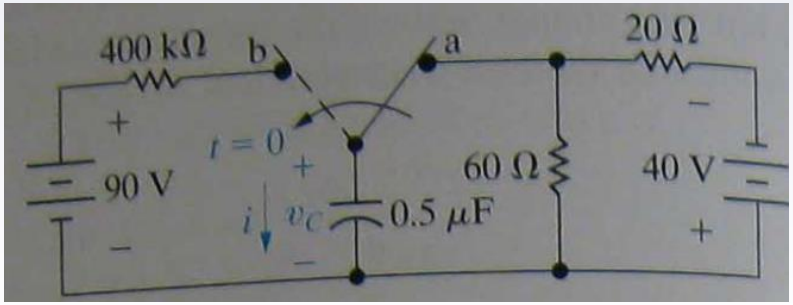
d) Substituting the appropriate values for v_f, $v(0)$ and t into the general formula

$$v_C(\mathrm{t}) = 90 + (-30 - 90)e^{-5t} = 90 - 120e^{-5t}\ V, \qquad t \geq 0$$

The Step and the Total Responses of RL and RC Circuits

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- Example-7: The switch in the given circuit has been in position a for a long time. At $t = 0$, the switch moves to position b.



e) What is the expression for $i(t)$ when $t \geq 0^+$?

Solution: e) The value for τ doesn't change. Thus we need to find only the initial and final value for the current in the capacitor. When obtaining the initial value, we must get the value of $i(0^+)$, because the current in the capacitor can change instantaneously. This current is equal to the current in the resistor, which from Ohm's law is

$$i(0^+) = \frac{[90 - (-30)]}{400 \times 10^3} = 300\ \mu A$$

The capacitor voltage cannot change instantaneously. The final value of $i(t) = 0$, so

$$i(t) = 0 + (300 - 0)e^{-5t} = 300e^{-5t}\ \mu A$$

The Step and the Total Responses of RL and RC Circuits

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- Example-7: The switch in the given circuit has been in position a for a long time. At $t = 0$, the switch moves to position b.

f) How long after the switch is in position b does the capacitor voltage equal zero?

Solution: f) We solve for the time when $v_C(t) = 0$

$$120e^{-5t} = 90 \quad \text{or} \quad e^{5t} = \frac{120}{90}$$

$$t = \frac{1}{5}\ln\left(\frac{4}{3}\right) = 57.54\,ms$$

Note that when $v_C = 0$, $i = 225\ \mu A$ and the voltage drop across the 400 kΩ resistor is 90 V.

Solution: f) We solve for the time when $v_C(t) = 0$

$120e^{-5t} = 90$ or $e^{5t} = \frac{120}{90}$

$t = \frac{1}{5}\ln\left(\frac{4}{3}\right) = 57.54\,ms$

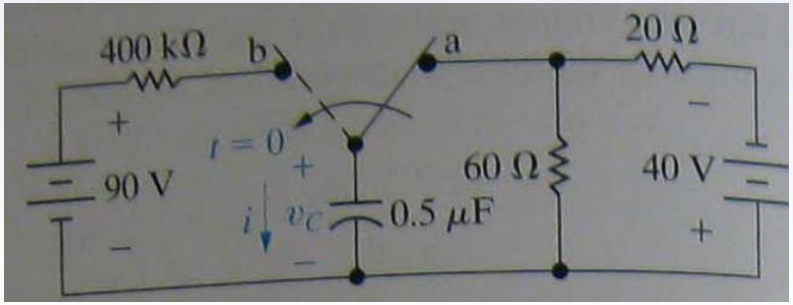
Note that when $v_C = 0$, $i = 225\ \mu A$ and the voltage drop across the 400 kΩ resistor is 90 V.

Sequential Switching

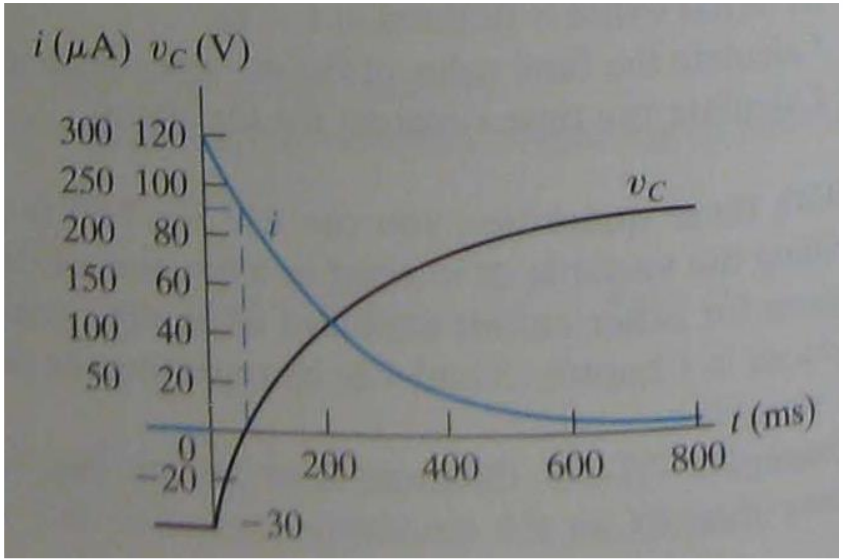
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- Example-7: The switch in the given circuit has been in position a for a long time. At $t = 0$, the switch moves to position b.

g) Plot $v_C(t)$ and $i(t)$ versus t.



Solution: g)



End of the Lesson (19.12.23)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist